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Phosphorus in rivers — ecology and management

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Abstract

This paper examines the nature of the risk to riverine ecosystems from artificially enhanced loads of phosphorus, considers the key sources of phosphorus enrichment and ways of controlling them, and provides a framework for developing control strategies. The aquatic plant community is the foundation for a healthy and diverse riverine ecosystem, providing food, shelter and breeding habitats for a wide range of animal species. Phosphorus enrichment in rivers can degrade the plant community by altering the competitive balance between different aquatic plant species, including both higher plants and algae. This has consequences for the whole ecosystem. To promote healthy riverine plant communities and the wide range of animal species dependent on them, phosphorus concentrations need to be reduced to as near to background levels as possible. The risk of adverse effects declines as phosphorus concentrations approach background levels, such that any incremental reduction should be seen as a positive step towards trophic restoration. Pragmatic management targets vary between 0.02 and 0.1 mg l^{-1} soluble reactive phosphorus, depending on river type, with an interim target of 0.2 mg l^{-1} for heavily enriched rivers. Continuous point sources of phosphorus, dominated by sewage treatment works, have a highly important influence on levels of bioavailable phosphorus in the water column through the growing season. It is important to tackle point sources comprehensively so that reductions in phosphorus concentrations are maximised during this critical time of year. Diffuse sources of phosphorus, particularly from agriculture, are a major contributor to phosphorus levels in riverine sediments, where it can be utilised by benthic algae and rooted plants. This phosphorus can also be released into the water column by a variety of processes. As point sources are brought under control, the relative contribution from diffuse sources becomes increasingly important. An integrated programme of control, involving proactive action on both point and diffuse sources, will be required in most circumstances to bring phosphorus levels in the water column and sediment down to near background levels. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Phosphorus; Eutrophication; Rivers; Nutrient control

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1. Introduction

Eutrophication is one of the most serious problems facing the ecology of freshwaters in the UK and elsewhere. The nutrient status of many lakes and rivers has increased dramatically over the past 50 years in response to increased collection and discharge of domestic wastes, increased loadings to collection systems (including the use of phosphate detergents), and widespread agricultural intensification.

There are statutory obligations and strong political pressures for greatly increased emphasis on the control of phosphorus levels in UK rivers (Table 1). It is important to note that phosphorus control in rivers is not only relevant to the ecology of the river itself, but also to any static waterbodies or wetlands that are fed by the river. The recently agreed EC Water Framework Directive provides an important new impetus for the general control of phosphorus, and also gives a strong focus on priority areas, particularly Special Areas of Conservation (SACs) under the EC 'Habitats' Directive, Sites of Special Scientific Interest (SSSIs) under the Wildlife and Countryside Act 1981, and 'sensitive areas' under the EC 'Urban Waste Water Treatment' (UWWT) Directive. In 1998, the government announced an extensive programme of improvements that it expects the water industry to undertake by 2005 (DETR, 1998), including measures to protect all of these types of priority areas from the adverse effects of effluent discharges to rivers (Environment Agency, 1998).

Some of the pressures in Table 1 are specific to phosphorus whilst others require the more general management of anthropogenic influences to protect river ecology. Amongst these anthropogenic influences, eutrophication is a primary consideration.

2. The nature of the risk to river ecology

A healthy submerged plant community is vital to the nature conservation value of a river. Apart from being of conservation importance in its own right, it provides essential resources for a range of animal species, associated with feeding, shelter and breeding. Dependent animals include invertebrates such as blackfly larvae, caddis-fly larvae and gastropod molluscs, which either filter-feed or graze algae off plant leaves and stems. Many fish species use submerged beds of higher plants as a spawning substrate, with the beds then becoming essential refuge and feeding habitat for juveniles. Bird species may graze plant shoots or feed on resident invertebrates, often utilising both food sources. Water-crowfoot (*Ranunculus* spp.) beds and associated plant and animal species are of particularly high nature conservation importance, being scheduled as a priority habitat under the EC Habitats Directive. Plant communities characteristic of other river types, such as the moss floras typical of many upland watercourses and the diverse plant associations of sluggish lowland rivers, are also of high nature conservation importance in terms of their contribution to the overall diversity of riverine plant assemblages nationally.

Of the major plant nutrients, phosphorus is typically in shortest supply in rivers and other freshwaters and so generally has the greatest potential to limit plant growth. Whether nitrogen or phosphorus availability actually limits plant growth in a river depends on many factors, including light intensity, current velocity and the availability of other major and trace plant nutrients. However, an assessment can be made of which of these two major nutrients may be limiting growth purely from consideration of their relative and absolute availabilities in the environment.

Fig. 1 plots the median N/P ratio (using total inorganic nitrogen, TIN, and soluble reactive phosphorus, SRP) against the median SRP concentration for over 5000 sites on rivers across England and Wales. It has been divided up into five zones on the basis of a threshold N/P ratio (broadly equating to the dividing line between phosphorus and nitrogen limitation), SRP concentrations (where the likelihood of phosphorus limitation of algal growth might be considered to change), and total inorganic nitrogen (TIN) concentrations (which are plotted as a function of the N/P ratio and SRP thresholds). The graph indicates that, at SRP concentrations at which phosphorus might conceivably be limiting growth, N/P

Table 1
Key obligations and pressures driving phosphorus control in rivers

Obligation/pressure	Relevant requirements
EC Directive establishing a framework for Community action in the field of water policy — the Water Framework Directive	To reach and/or maintain at least 'good' ecological status in all surface waters, and to comply with any standards/objectives for EU 'Protected Areas'[a], by the year 2010. The assessment of ecological status includes specific consideration of nutrient status.
EC Directive 91/271/EEC concerning urban waste Water treatment — the Urban Waste Water Treatment (UWWT) Directive	To identify 'sensitive areas' and to install appropriate treatment facilities for phosphorus and/or nitrogen removal at sewage treatment works serving more than 10 000 'population equivalents'.
EU Directive 92/43/EEC on the conservation of Natural habitats and of wild fauna and flora — the Habitats Directive	To maintain or restore the 'favourable conservation status' of specified natural habitats and species of wild fauna and flora, mainly through a network of Special Areas of Conservation and using the precautionary approach as necessary.
EU Directive 79/409/EEC on the conservation of Wild birds — the Birds Directive	To protect and, where necessary, enhance populations of specified bird species (including many wetland species), through a range of measures including the designation of Special Protection Areas and the control of pollution.
The 1981 Wildlife and Countryside Act	To protect the national network of Sites of Special Scientific Interest as representative examples of the habitats, flora and fauna of the UK. The network includes a series of riverine SSSIs in England and Wales.
The Rio Convention on Biological Diversity	Through the UK Biodiversity Action Plan, to conserve and where possible enhance populations and natural ranges of priority native species, the quality and range of priority habitats, and the biodiversity of habitats where this has been degraded.
UK Government programme of improvements to the Aquatic environment (DETR, 1998)	The water industry to implement a range of improvement measures between 2000 and 2005, including the removal of phosphorus from a range of sewage effluents discharging into specified riverine SSSIs or rivers feeding wetland SSSIs.

Note: Each EU Directive is brought into UK law by statutory 'Regulations' which have not been detailed here.
[a]'Protected Areas' include those designated under the EC UWWT, Habitats and Birds Directives.

ratios are nearly always above eight and typically well above this value. This means that nitrogen is invariably surplus to plant requirements relative to phosphorus in such situations.

Increasing phosphorus availability can affect the growth rates and standing crop of individual plant species (algae and higher plants) and, consequently, the competitive balance within riverine plant communities. Such changes can have knock-on consequences for animal species dependent on

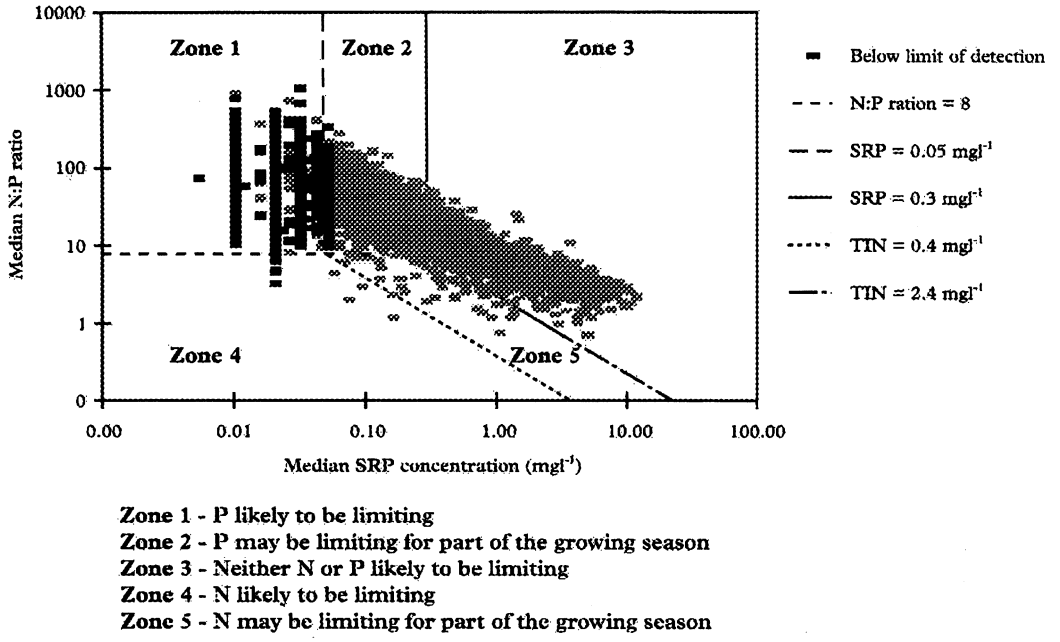


Fig. 1. The potential for phosphorus and nitrogen limitation of riverine plant growth in England and Wales (modified from Mainstone et al., 1995).

riverine plants. In addition, changes in the plant community can lead to severe nocturnal sags in dissolved oxygen (due to plant respiration) that stress the most sensitive animal species and may result in mortality. In upland streams and rivers, the entrapment of fine particles by excessive growths of epilithic algae exacerbates siltation problems, leading to spawning difficulties for species such as Atlantic salmon (*Salmo salar*).

There are four principal ways in which elevated phosphorus levels can affect riverine plant communities:

1. by increasing higher plant growth rates and thereby creating a large standing stock that regrows rapidly following management;
2. by the encouragement of higher plant species whose growth rates are geared to higher nutrient levels, thereby altering species composition/balance;
3. by the encouragement of epiphytic, epibenthic, filamentous and planktonic algae, thereby reducing the amount of light reaching the leaves and stems of higher plants and interfering with the success of seed germination and seedling growth; and
4. by reducing rooting depth and thereby making higher plants more susceptible to being ripped out of the substrate under high river flows.

All of these mechanisms are probably dependent upon phosphorus levels in both the sediment and the water column. For any of these mechanisms to operate in response to artificially enhanced phosphorus concentrations, background levels of phosphorus in the river have to lie below concentrations that trigger an effect. If this is not the case, no effect of increasing phosphorus levels above background concentrations can be expected. The threshold concentration for effects will vary between mechanisms, environmental compartments (i.e. the water column and the substrate) and river types.

Little detailed work has been undertaken on the role of phosphorus in the competition between different higher plant species, but somewhat more is known about the risks from mechanism (3) due

to extensive work on algal growth rates at varying levels of ambient soluble phosphorus (e.g. Reynolds, 1984). The half-saturation coefficient, *Ks*, is often used to describe the response of individual algal species to nutrient availability (Kilham and Hecky, 1988) — it is defined as the external nutrient concentration at which half of the maximum growth rate is achieved. *Ks* values are often cited to support the idea that the growth rate of the algal community as a whole is only limited at phosphorus concentrations of < 10 μg l^{-1} SRP. Reported *Ks* values for individual species range between 1 and 364 μg SRP l^{-1} (Reynolds, 1984), such that at any concentration within this range, some species will be growing at their maximal rate. However, it is self-evident that the growth of individual species may be limited at concentrations substantially greater than 10 μg l^{-1}. Substantial increases in growth are also possible above the *Ks* value of a species.

Algae which dominate at high phosphorus concentrations appear to have high *Ks* values and are out-competed at lower concentrations by algae with low *Ks* values. They are sometimes, but not always, faster-growing than those which dominate at lower concentrations. In any case, higher algal standing crops may be achieved by such algae by a number of different methods, such as:

- being less palatable, so grazing losses are lower;
- being physically stronger or more streamlined, so scouring losses during high flows are lower; or
- being adapted to photosynthesise at lower light intensities, so self-shading does not become limiting until higher standing crops are reached.

The main point from such work is that phosphorus has the potential to greatly affect the growth rates and standing crop of riverine algal communities at water column concentrations up to at least 0.2–0.3 mg l^{-1}. Artificial enrichment to such levels is, consequently, extremely important in the battle for dominance within the plant community. Current thinking on competition between algal and rooted plant species in lakes (e.g. Timms and Moss, 1984; Scheffer et al., 1992; Moss, 1997) suggests that the influence of phosphorus in freshwaters extends over a wider range of concentrations, up to and probably beyond 1 mg l^{-1}.

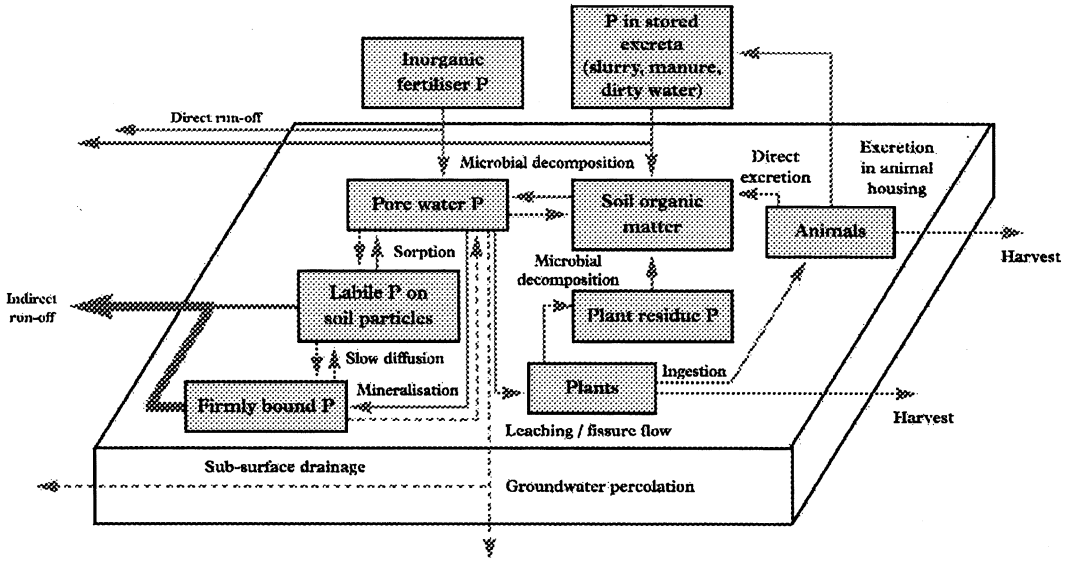


Fig. 2. Phosphorus behaviour in soils and pathways to river (after Mainstone et al., 1996).

3. General behaviour of phosphorus in river catchments

Phosphorus is delivered to the river system from a range of sources, varying in its bioavailability from source to source. The load from point sources (which are dominated by sewage treatment works) is typically highly bioavailable and is delivered along with considerable loads of readily degradable organic material. The delivery of diffuse loads is more complicated (Fig. 2) and highly seasonal, but is largely dictated by the strong affinity of phosphorus for particulates. The majority of the diffuse annual load is, therefore, generally delivered in surface run-off attached to soil particles. However, much higher proportions of soluble phosphorus occur when livestock excreta (in forms such as slurry and 'dirty water') or soluble inorganic fertilisers are washed off the land soon after application.

Sub-surface drainage and leaching may be important pathways under certain conditions, particularly if the soil is overloaded with phosphorus. Sandy soils and underlying sandstone geology are particularly vulnerable since they have a very low adsorption capacity for phosphorus. Other soils and geologies are less vulnerable, but may be more at risk than supposed due to macropore and fissure flow within the soil/rock structure (see Heathwaite, 1997 for more detail on phosphorus losses from soil to water).

Once in the river, phosphorus is highly chemically and biologically active, undergoing numerous transformations and moving between the particulate and dissolved phases, between the sediment and water column, and between the biota and abiotic environment (Fig. 3). Physical deposition and resuspension of particulates are obvious methods of phosphorus transfer between the water column and bed sediments, but direct adsorption/desorption processes between the two compartments are also important and will depend upon the equilibrium phosphate concentration of the sediment (EPC, the SRP concentration in the water column or pore water that produces no net flux between the particulate and dissolved

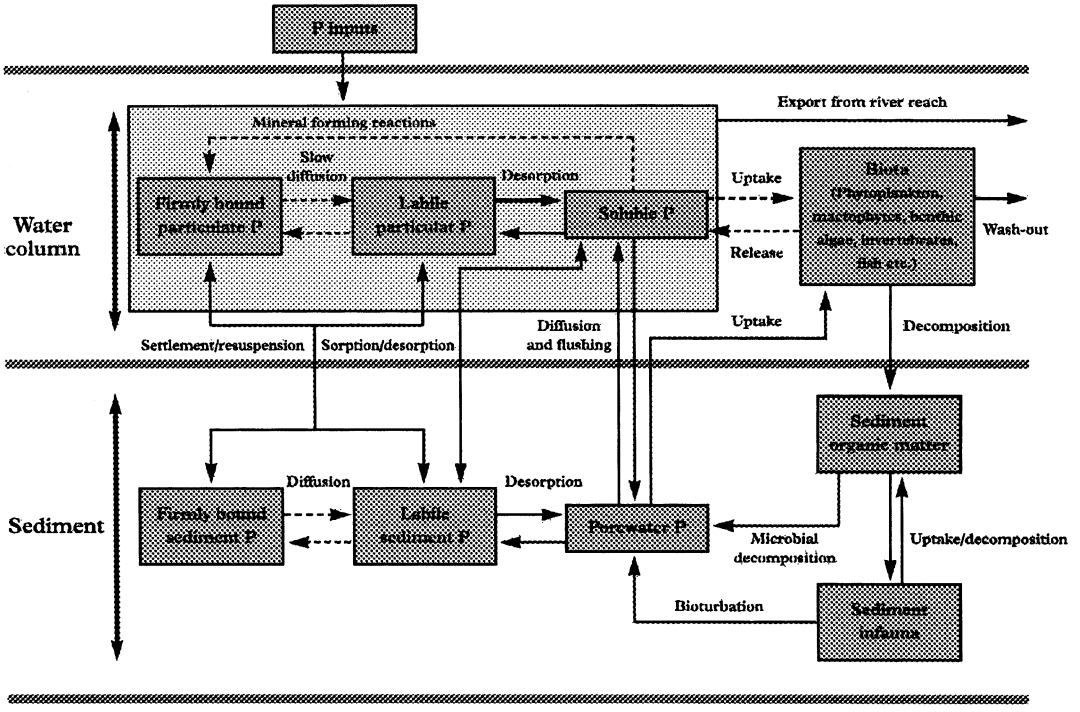


Fig. 3. Phosphorus behaviour in rivers (after Mainstone et al., 1996).

phase — see House et al., 1995), SRP levels in the overlying water, and current velocity (the latter dictating the sharpness of the diffusion gradient). SRP may be incorporated into calcium carbonate precipitates in hard-water areas, but this seems to occur at the water–sediment interface rather than in the water column, and both inorganic and organic phosphorus can be bound into biofilms at this interface (Neal, 2002a,b; Jarvie et al., 2002a).

Labile phosphorus attached to suspended particulates can rapidly desorb into the water column and become bioavailable, again depending upon the EPC of the particles and the SRP concentration in the water column. Firmly held phosphorus deep within the particle matrix can diffuse slowly into the water column, which is most likely to be an important mechanism for the river once particulates have settled as bed sediments. Soluble phosphorus can be incorporated into inorganic phosphate minerals by precipitation, particularly in association with calcium (in hardwater rivers), iron and aluminium (in softwater rivers). The precipitation of soluble phosphorus with calcium is particularly likely to occur below sewage treatment works in rivers with calcareous waters (e.g. House and Denison, 1997), where both calcium and soluble phosphorus concentrations are very high. Colloids of calcium phosphate minerals can be generated in the water column, whilst algal biofilms are thought to be involved in the coprecipitation of calcite and phosphorus onto bed sediments and plants (Hartley et al., 1997). However, in various UK studies, calcium co-precipitation has not been found to produce phosphorus removal rates that would amount to environmental self-regulation of phosphorus levels.

Rooted aquatic plants are capable of deriving much, if not all, of their phosphorus requirements from the sediment, but there is considerable debate about the relative importance of root (via the sediment) and shoot (via the water column) uptake in real systems. Some studies have suggested that the uptake of phosphorus directly from the water column can be very important (Robach et al., 1995, 1996), particularly when SRP concentrations in the overlying water are higher than in sediment pore waters (Pelton et al., 1998). However, contrary to some beliefs, shoot uptake appears to have the potential to be important even at low water column concentrations of SRP (Pelton et al., 1998). Filamentous, epiphytic and planktonic algae generally take phosphorus directly from the water column by necessity, although benthic algae (including filamentous mats) will utilise both sources and epiphytic algae can derive some nutrition from the host plant. Microbial uptake from the water column and more particularly within the sediment can be substantial. Decay of plant shoots and the mineralisation of organic matter by the microbial community will lead to phosphorus release into both sediment pore waters and the water column, offset to varying degrees by uptake by higher plants and algae.

The combination of all of these processes, in tandem with variations in river flow and other environmental factors (such as temperature), leads to a strong seasonality in phosphorus behaviour. Phosphorus will tend to be taken up and retained in the sediment and the biota (higher plants, algae and bacteria) through the summer months, whilst much of the accumulated load will be scoured and sorbed out during the high flows of autumn and winter (depending upon the strength of winter flows). Strong microbial activity in the spring can produce a significant flush of phosphorus from the sediment in advance of strong biological uptake through the growing season. Seasonal patterns in the phosphorus loads from different sources add a further layer of complexity.

Inevitably, the importance of different processes will vary greatly between river reaches. In very swift-flowing river reaches, substrates will be coarse and areas of sediment deposition restricted, leading to limited accumulation of phosphorus within the sediment over the summer months. The high scouring velocities are unfavourable to plant growth and much water column SRP goes unutilised down the river. This said, the widespread observations of siltation problems in coarse riverine substrates in many UK rivers suggest that accumulation of phosphorus may be more important than anticipated. In

sluggish river reaches, the fine sediments and low bed velocities create ideal conditions for the accumulation of phosphorus and the potential for subsequent biological utilisation. Populations of higher plants and planktonic, epiphytic, benthic and/or filamentous algae can strip SRP out of the water column to leave very low levels through the summer months, even though the river is highly enriched.

4. Environmental factors confounding the effects of phosphorus

There are numerous other environmental factors that influence the growth and composition of higher plant communities in rivers. Some of these are influences that are generally not altered by human activities (such as catchment geology and morphology), whilst others can be greatly modified by man. Community composition can, therefore, be seen as strongly determined by fundamental, fixed factors, and modified to a greater or lesser extent by anthropogenic impacts on a range of secondary factors, one of which is phosphorus levels.

Table 2 summarises key secondary factors that can influence plant communities and thereby serve to obscure the role of phosphorus in rivers. The important point to bear in mind is that elevated phosphorus concentrations are rarely the only important anthropogenic influence on the plant community in a given river reach. Only when all other influencing factors can be held constant whilst phosphorus availability changes within ecologically relevant ranges can the effects of phosphorus be unequivocally observed. In real systems, this is rarely the case, and the impact of phosphorus is difficult to separate from other types of impact. This must not be taken as a justification for ignoring the role of phosphorus, which can be thought of as setting the underlying potential for impact upon which other anthropogenic factors are superimposed.

5. Natural, background and existing levels of phosphorus in rivers

In pristine rivers, annual loads of phosphorus to the river system are extremely low, since phosphorus is not abundantly available from the majority of natural geologies. The natural nutrient status of the watercourse is largely dependent upon nutrient retention in the system and subsequent spiralling downstream, as nutrients are continually taken up and released by the abiotic and biotic mechanisms described above. Through the summer months, SRP levels in the water column are dictated by the concentrations in baseflow, as modified by instream processes which typically result in some of this SRP being stripped out of the water column. The release of phosphorus from riverine sediments under pristine conditions mainly occurs during autumn flushing events, when large amounts of particulates are resuspended.

There is no argument that natural levels of SRP in upland UK rivers are very low (below 10 $\mu g\ l^{-1}$), since there are plenty of examples where

Table 2
Modifiable environmental factors that confound the effects of phosphorus on submerged higher plants in rivers

Factor/key mechanism of impact	Reduced flows	Channel modifications	Pt. source discharges	Diffuse pollution	Loss of tree cover
Substrate (siltation)	*	*	*	*	
Current velocity	*	*			
Organic status			*	*	
Turbidity		*	*	*	
External shading					*
Residence time	*	*			
Other trace nutrients			*	*	
Algal/higher plant grazing	*	*	*	*	

such concentrations can still be found. There is much more debate about what levels might naturally be expected in lowland rivers types, where human activities have intensified in the UK to the extent that near-pristine examples do not exist. However, there is no evidence to suspect that the natural situation in lowland rivers differs from the model described above, with low external inputs and a focus on nutrient retention and spiralling. For instance, work by researchers of unpolluted calcareous streams (such as Robach et al., 1996 and Mulholland and Hill, 1997) indicate that the very low natural concentrations of SRP in calcareous aquifers give rise to natural riverine concentrations of less than 10 $\mu g\ l^{-1}$. This is supported by observations of background SRP concentrations of less than 10 $\mu g\ l^{-1}$ in chalk aquifers of southern England (Mainstone et al., 1999a). Phosphorus accumulation within the system may result in slightly higher natural concentrations in the lower reaches of large rivers, due to internal cycling, but this is unlikely to produce levels greater than 30 μg SRP l^{-1}.

Whilst sandstone geologies are more susceptible to contamination of baseflow and run-off and may, therefore, have suffered enrichment earlier than chalk rivers in the UK, it is likely that natural concentrations would not be any greater. Clay-based catchments may produce somewhat higher natural concentrations of total phosphorus than sand- and chalk-based catchments, due to the dominance of run-off as a hydrological pathway, but much of the natural load would be in non-labile particulate form and would not be biologically available (at least in the short term).

Background levels are distinct from natural levels in that they admit a certain amount of anthropogenic impact, but at an intensity which is judged unlikely to produce harmful effects. Typically, background levels are estimated for a specified reference or baseline date, before most of the adverse ecological impacts were visible in UK surface waters. Restoring phosphorus concentrations to this baseline date is then assumed to eliminate any ecological problems relating to phosphorus. A range of baseline dates has been employed in different studies, but pre-World War II (1930s) is the time most often used (prior to large-scale increases in loads from agricultural and domestic sources). The problem is that some rivers inevitably suffered enrichment problems at this time, such that the use of a reference date from this period on a particular catchment can produce estimates of phosphorus levels that are inappropriate from a management perspective.

Unfortunately, there are no recorded phosphorus concentrations in rivers from any candidate reference dates, for the estimation of either natural or background concentrations, and so the assessment of natural or background levels in an impacted catchment, therefore, has to rely on indirect methods. Assessment may be based on comparison with relatively unimpacted examples of the river type (possible in upland areas but generally difficult in lowland areas), hind-casting modelling (whereby the load entering the river network at a given reference date is estimated through the use of export coefficients and data on human activities within the catchment) or expert judgement. The estimation of reference levels of phosphorus (which may be natural or background depending on the policy adopted) is likely to be a cornerstone of the European strategy for controlling freshwater eutrophication and will be an important component of the assessment of ecological quality under the Water Framework Directive.

Background rates of phosphorus export to rivers vary widely, partly due to methodological differences and partly due to natural differences in catchment conditions. This leads to uncertainty over how to define reasonable target levels for management purposes. The majority of estimates range between 0.05 and 0.65 kg TP ha^{-1} $year^{-1}$ (e.g. Vighi and Chiaudani, 1985, Billen et al., 1991), and Table 3 shows how this range of export rates broadly equate to phosphorus concentrations in rivers of different characteristics (by dividing the catchment export rate by the annual stream flow). Export rates judged to be most relevant to different river types are shaded, producing estimated mean natural concentrations of below 30 μg TP l^{-1} for all case studies. Higher export rates are also shaded that yield riverine concentrations deemed unlikely to increase ecological risk greatly, as an indication of ecologically

Table 3
Estimated mean total phosphorus concentrations ($\mu g\ l^{-1}$) that would occur in selected UK rivers at a range of possible background export rates

	Afon Glaslyn @ Beddgelert	River Tees @ Middleton	River Otter @ Dotton	River Nadder @ Wilton
NGR	SH 592478	NY 950250	SY 087885	SU 098308
Mean annual rainfall	3067 mm	1553 mm	985 mm	906 mm
Mean annual runoff	2670 mm	1167 mm	483 mm	406 mm
Catchment area	68.6 km^2	242.1 km^2	202.5 km^2	220.6 km^2
Surface geology	Igneous	Sandstone/ limestone/ millstone grit	Sandstone	Chalk/ greensand
EC = 0.07 kg TP ha^{-1} $year^{-1}$	*2.6*	6.0	*14.5*	17.2
EC = 0.1 kg TP ha^{-1} $year^{-1}$	*3.7*	*8.6*	*20.7*	*24.6*
EC = 0.2 kg TP ha^{-1} $year^{-1}$	**7.5**	*17.1*	**41.4**	**49.3**
EC = 0.31 kg TP ha^{-1} $year^{-1}$	11.6	**26.6**	64.2	76.4
EC = 0.65 kg TP ha^{-1} $year^{-1}$	24.3	55.7	134.6	160.1

EC = Export coefficient; Italic type = most likely natural concentrations; Bold type = ecologically desirable background concentrations.

desirable background loads. Note that these concentrations are in terms of total phosphorus — SRP concentrations would be appreciably lower in natural systems.

It is enlightening to compare likely natural and background concentrations of SRP in UK rivers with existing levels. Fig. 4 shows the extreme differences in existing SRP concentrations between coarse categories of river in England and Wales, based on catchment altitude and river

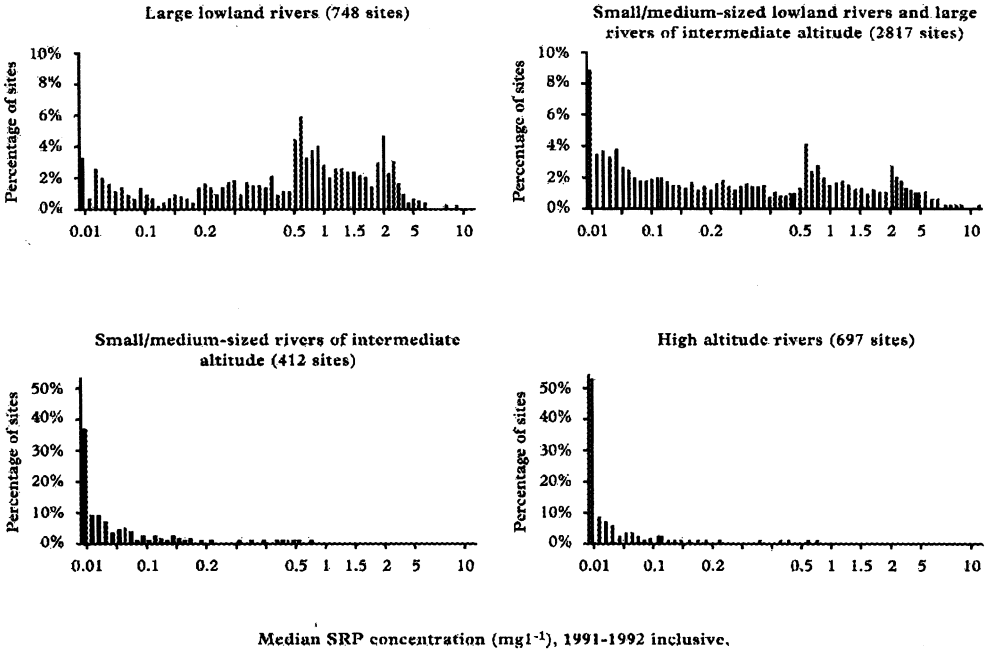


Fig. 4. Ambient phosphorus concentrations in different types of river in England and Wales (after Mainstone et al., 1995).

size. The majority of rivers with high altitude catchments still maintain median concentrations of below 0.03 mg l^{-1} (with approx. half below 0.01 mg l^{-1}), although some sites exhibit concentrations an order of magnitude higher. Upland rivers are highly sensitive to small increases in phosphorus availability, so even enrichment of the order of 10 μg l^{-1} is of concern. There are, therefore, significant numbers of upland rivers exhibiting worryingly high phosphorus levels. The picture is similar for smaller rivers at intermediate altitudes, but is very different for lowland rivers where a massive range of ambient SRP concentrations is found. Very low concentrations still occur in some lowland rivers, particularly the smaller ones, but the majority of large lowland rivers exhibit concentrations of over 0.3 mg l^{-1}, with many exceeding median concentrations of 1 mg l^{-1}. This represents widespread and heavy enrichment.

6. Establishing targets for management

To promote healthy riverine plant communities and the wide range of animal species dependent on them, phosphorus concentrations should be reduced to as near background levels as possible. The risk of adverse effects due to phosphorus declines as phosphorus concentrations approach background levels, such that any incremental reduction should be seen as a positive step towards trophic restoration. Given the sliding scale of ecological risk associated with riverine phosphorus concentrations, and the numerous confounding factors operating, it is unrealistic to expect to identify particular threshold concentrations of phosphorus that will automatically safeguard the plant and animal communities of different river types. However, it is clear that the level of risk changes most rapidly over the concentration range from natural conditions to approximately 200–300 μg l^{-1} (see previous sections). A pragmatic approach to setting targets is, therefore, required, involving the identification of phosphorus levels within this concentration range that are achievable and that approach those expected at low levels of anthropogenic input for the type of river in question.

The Environment Agency's national strategy for eutrophication control (Environment Agency, 2000) includes a family of phosphorus targets for rivers (Table 4). The Agency's intention is that target selection for any given river will depend upon site-specific considerations and may be chosen on the basis of historical conditions or on conditions in similar but less impacted examples of the river type, together with consideration of the feasibility of reducing phosphorus loads in the light of the nature of anthropogenic stresses. For the purposes of protecting priority wildlife sites (riverine SACs and SSSIs), the targets are tentatively linked to their likely generic suitability for different river types, as a coarse guide to what might be appropriate in any given situation. Inevitably, where a river supplies a standing water requiring trophic restoration, riverine targets will need to consider the desired phosphorus concentrations in that standing water. The setting of interim targets should also be considered for highly enriched rivers to provide greater encouragement to restoration efforts and to assist in the reporting of real progress towards an ultimate goal.

Table 4
Target phosphorus concentrations for rivers in England and Wales (Environment Agency, 2000), with suggested applications to different river types

Target	Mean SRP (mg l^{-1})	Suggested application
1	0.02	Upland watercourses and headwaters generally.
2	0.06	Rivers on chalk, hard sandstone and limestone.
3	0.1	Lowland rivers on clay and alluvium.
4	0.2	Interim target for heavily enriched rivers.

Water column concentrations are inevitably only one facet of riverine enrichment (although they are generally a useful indicator of enrichment in other environmental compartments) and greater attention needs to be focused on the status of the sediment in future. Sediment phosphorus targets for different river types, probably based upon the equilibrium phosphate concentration, ultimately need to be developed to provide a holistic framework for phosphorus control.

7. The importance of different phosphorus sources

Estimating the contribution of different sources to the total phosphorus load entering a receiving water (be it a river, lake or coastal water) is typically performed using annual estimates of total phosphorus. This is sensible for rapid evaluation, since there is great uncertainty in the estimation of phosphorus bioavailability and some difficulties in quantifying the seasonality of inputs. Whilst such estimates are generally adequate for lakes, where the whole of the phosphorus load is potentially relevant, they can be misleading for rivers, where much of the total load can be exported from the river soon after it has entered. For this reason, it is important to understand the nature and timing of different sources of phosphorus and how they are likely to impact upon the growth of higher plants and algae in the river.

Relatively simple methods of apportioning the total instream load between different sources, such as the subtraction of point source loads from total river loads, the interpretation of flow–concentration relationships and the use of hydrograph separation, can be useful. Simple export coefficient approaches can be taken further in complex models such as HSPF and CREAMS (Mainstone et al., 1996), but these require detailed datasets and are expensive to run. GIS-based models of phosphorus export, such as AGNPS (Morse et al., 1994), provide a compromise between complex models and simple spreadsheet calculations. Such models can tackle the issue of seasonality in diffuse source loads by simply apportioning the estimated annual load though the year according to rainfall patterns, and can generate phosphorus concentrations in the river

Table 5
Examples of annual phosphorus budgets (in tonnes P per year)

Source	Upper reaches of the Hampshire Avon (Parr et al., 1998)	Warwickshire Avon (Iversen et al., 1997)	Pevensey Levels (Parr and Mainstone, 1997)	River Ant (Johnes et al., 1994)
Atmospheric/natural	12.5 (14.0%)	57.9 (5.5%)	0.6 (1.6%)	0.08 (1.6%)
Inorganic fertiliser	19.9 (22.4%)	209.5 (20.0%)	2.5 (7.4%)	1.04 (21.3%)
Livestock	18.7 (21.0%)	99.5 (9.5%)	2.4 (7.0%)	2.89 (59.3%)
STWs	35.5 (39.9%)	654.3 (62.6%)	28.7 (84.0%)	0.86 (17.6%)
Unsewered population	23.8 (2.3%)	–	–	–
Industry	2.5[a] (2.8%)	–	–	–
Total	89.1 (100%)	1045.0 (100%)	34.2 (100%)	4.87 (100%)
Catchment area (km^2)	1249	2892	56	49.3
P load exported to river ($kg\ ha^{-1}\ year^{-1}$)	0.7	3.6	6.1	1.0

[a] Fish farming.

through the year and across the catchment. This comprehensive consideration of diffuse source loads is an important improvement on point source-based dilution models such as SIMCAT, providing a more realistic picture of the contribution of different sources to riverine phosphorus levels and giving a valuable assessment of the geographical distribution and source of diffuse loads.

In simple terms, the total load to the river can be broadly divided into point source inputs and diffuse inputs. Point sources are typically dominated by sewage treatment work (STW) effluents (although industrial effluents can be phosphorus-rich and should always be checked), whilst diffuse sources are dominated by agriculture (other diffuse sources include run-off from roads and urban areas, septic tanks and forestry plantations). Diffuse agricultural sources contribute a substantial part of the annual load to rivers in all but the most urban catchments (some examples are provided in Table 5), but the ecological relevance to the river of much of this load from this source may be low. Since phosphorus absorbs strongly onto soil particles, most of the diffuse load enters the river during run-off events in autumn and winter, under conditions of high river flow. This means that much of the load from this source will be immediately flushed out of the system along with a large proportion of the phosphorus accumulated in the river over the course of the summer.

This is not to say that diffuse loads are unimportant to the river, since phosphorus-rich soil particles retained in the river system (in areas of sufficiently low hydraulic energy to permit settlement) can supply plant growth in subsequent growing seasons, either directly via root uptake or via subsequent release (internal loading) to the water column. The scale of retention is greater in years of low winter flows, due to the reduced flushing rates, creating greater scope for impact in the following year. There is also potential for phosphorus applied to the land to leach into sub-surface drainage and groundwater where soils are heavily overloaded with phosphorus. If this occurs, diffuse sources start to become a significant source of input throughout the year. Diffuse loads are also highly important to any lakes and wetlands fed by the river, where retention is generally much higher (and the entire annual load is therefore of relevance). It is, therefore, vital that diffuse sources are effectively controlled, but it is equally important to place their contribution to riverine eutrophication into context by qualifying the results of annual budgets.

Point sources are more important to riverine phosphorus concentrations than estimates of annual total phosphorus loads suggest, since they enter the river continually through the year and are at minimum dilution through the growing season when phosphorus concentrations in the water column are critical. In addition, the phosphorus load in STW effluents is highly bioavailable and can, therefore, have an immediate impact. Detailed studies have demonstrated the primary importance of dilution of point source loads to the water column concentrations of soluble phosphorus in many rivers (e.g. House and Denison, 1997; Neal et al., 2000a,b), even in chalkstreams where chemical precipitation with calcium carbonate might be anticipated to regulate phosphorus availability to some extent.

Fig. 5 provides a stylised illustration of the seasonality in point source and diffuse loads, and gives an indication of how the contribution from each type of source changes with flow through the year. Inevitably, there will be large variation about this general pattern, mainly depending upon the degree of urbanisation of the catchment. The potential contribution of point sources to the sedimentary phosphorus reservoir should also not be ignored (Chambers and Prepas, 1994), with inputs under low-flow conditions permitting enhanced rates of accumulation of the highly organic particulate load (particularly if the growth rates of higher plants have been stimulated by the effluent, causing enhanced trapping of particulates).

8. The need for integrated control of phosphorus sources

If phosphorus concentrations in UK rivers are to be reduced to envisaged target levels, it is clear

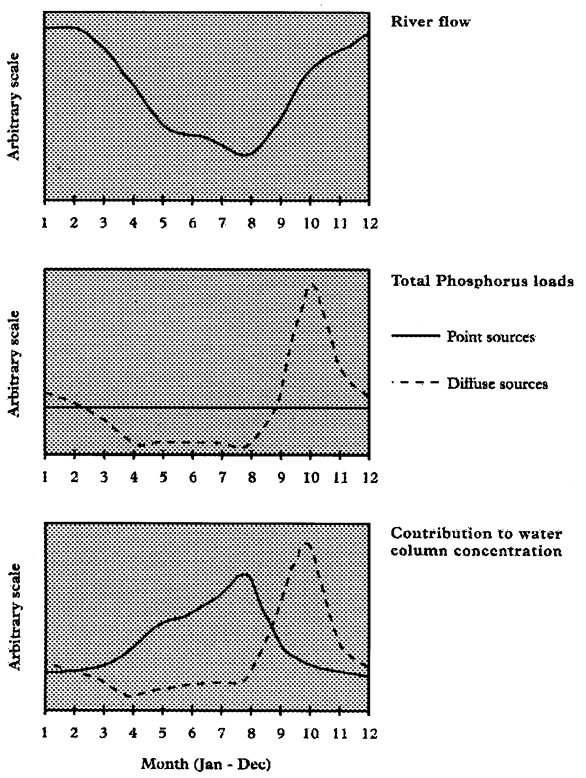


Fig. 5. Typical seasonality in the contribution of point and diffuse sources to phosphorus concentrations in the river.

that an integrated approach is required to control loads, involve proactive action on both point and diffuse sources and manage the internal cycling of phosphorus where necessary. Within any given catchment, the importance attached to controlling point and diffuse sources will vary depending on population density, land use intensity and the nature of the river. However, there is no doubt that point sources are far easier to control than diffuse sources, and that, given the nature and timing of point source loads outlined above, a good return for the resources invested is likely to accrue from tackling point sources comprehensively. In any given situation, the dominant contribution to point source loads from STWs may be significantly augmented by industrial and fish farm discharges, which, therefore, need to be considered fully in any control programme (see Mainstone et al., 1996 for more detail). Diffuse sources generally attain their greatest importance in river reaches where the retention of particulates from run-off is highest (i.e. in sluggish, silty sections or in sections dominated by coarse substrates that are subject to excessive siltation).

9. Control of diffuse sources

Whilst forestry, urban drainage and the unsewered population can be important in different situations, agriculture typically dominates the diffuse source contribution. A range of best management practices can be employed to reduce loads from livestock and arable agriculture (Mainstone et al., 1996). Agricultural controls can be divided into three main areas (Fig. 6), comprising minimisation of inputs, retention of phosphorus in the soil, and run-off capture in buffer zones. Proper consideration of all three areas, with an emphasis on input minimisation and sustainable soil management (including erosion prevention) is the key to successful diffuse source control. A heavy reliance on run-off amelioration by buffer zones, which can easily be circumvented by sub-surface hydrological pathways (enhanced by under-drainage systems), break-through surface run-off and artificial run-off pathways (paths, tracks and roads), does not constitute a comprehensive solution. However, judicious positioning, design and maintenance of buffer areas at key points within the catchment (not just immediately adjacent to watercourses) is an important component of an integrated control strategy.

Considerable change is required to current phosphorus application regimes in agriculture, and this issue has important links to the ultimate sustainability of phosphorus stripping programmes for point sources. It is important to note that the concentrations of extractable phosphorus are very high in many agricultural soils in England and Wales, and in many lowland soils, they are much higher than required to insure against crop growth limitations (Mainstone et al., 1996). This leads to unnecessarily high concentrations of phosphorus in run-off and the possibility of leaching into sub-surface drainage or groundwater. The Code of Good Agricultural Practice for the Protection of Water (MAFF, 1998) advises farmers to

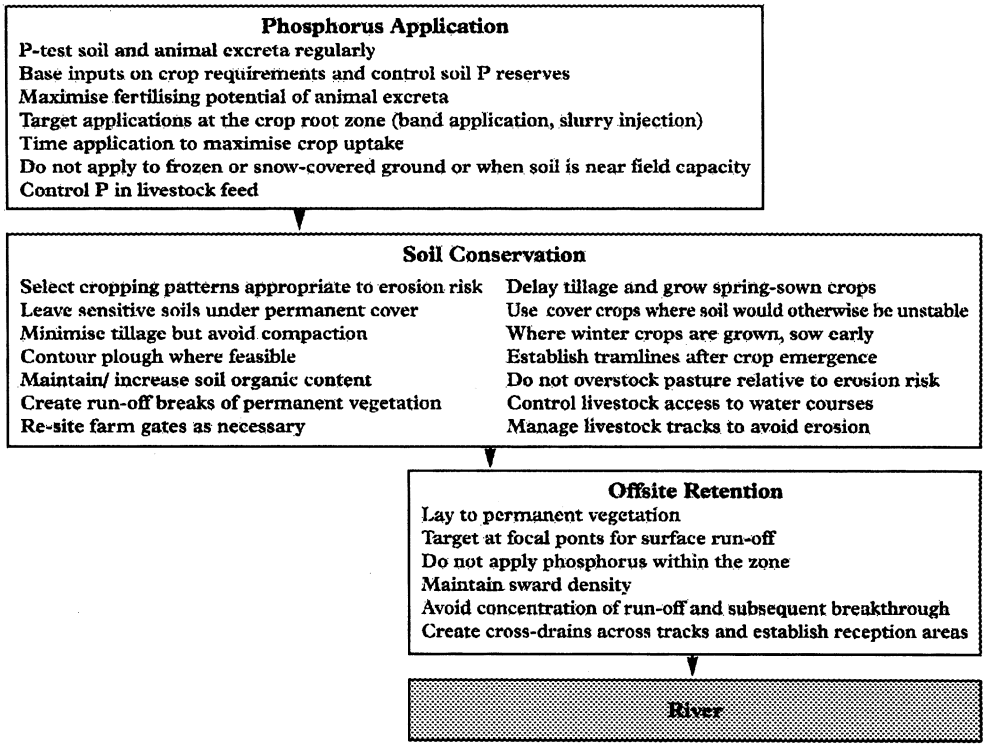


Fig. 6. Management practices to reduce diffuse agricultural loads of phosphorus to rivers (after Mainstone et al., 1996).

limit phosphorus applications based on soil phosphorus reserves, which will hopefully help to reduce the potential for diffuse phosphorus losses. However, this guidance is not widely read by farmers and in any case will not help to bring about a reduction in soil phosphorus levels in areas of intense slurry spreading activity, where sizeable applications to land are still possible under the Code even where there is no nutritional requirement for phosphorus.

Erosion control is key to reducing phosphorus loads from agricultural land, since the majority of the diffuse phosphorus load is typically associated with particulates in run-off. The importance of maximising soil stability cannot be overstressed, which may involve leaving critical areas of the catchment under permanent vegetation (pasture or woodland) or modifying tillage regimes. Run-off breaks in large sloping fields, which may take the form of hedges but could be grassed banks, are vital in preventing the gradual migration of fine particulates towards the river network (Parr et al., 1999). Artificial run-off pathways can greatly alter the classic perceptions of run-off patterns and can necessitate catchment walk-overs to ensure that the right areas of land within the catchment are being targeted for attention. New guidance on erosion control has recently been produced by MAFF (1999), which should help to reduce particulate loads to rivers if promoted properly.

The mechanisms for introducing desired changes into agricultural management regimes are varied, ranging from advice and education, through financial support to regulation. Recent local initiatives (see Mainstone, 1999b) suggest that a great deal can be achieved by proactively raising awareness within the farming community, explaining the links between environmental problems in rivers and certain agricultural practices. A

detailed explanation of the agroeconomic benefits of changing practices is equally important and is being built into new advice on best management practices from the Environment Agency. Nutrient management is one area of agricultural activity where both the farmer and the riverine environment can benefit greatly by modifying management regimes. However, there are changes required by modern agricultural regimes that are highly unlikely to be brought about by advice and education (such as the conversion/reversion of critical areas to permanent vegetation), and in such cases, it is crucial that mechanisms are available to bring such changes about.

Agricultural grant aid schemes must be sufficiently flexible to incorporate the specific requirements of the aquatic environment — at present there is no grant aid available for specific measures to combat diffuse agricultural pollution. Recent reform of the EU Common Agricultural Policy, allowing environmental conditions to be attached to production subsidies, may provide useful leverage if implemented effectively in the UK. A range of possible regulatory scenarios also

Table 6
Advantages and disadvantages of chemical and biological phosphorus stripping (after Day and Cooper, 1992)

Advantages	Disadvantages
Chemical P removal	
Reliable, well-documented technique	Chemical costs higher than for biological systems.
Chemical costs can be reduced substantially if waste pickle[a] liquors (ferrous chloride or ferrous sulphate) are available and can be used	Significantly more sludge produced than waste water treatment process without metal addition; may overload existing sludge handling equipment; higher sludge treatment and disposal costs.
Controls are simple and straightforward — easy to maintain high P removal efficiency by controlling metal salt dosing rate.	Sludge does not dewater as well or as easily as conventional STW sludges where metal salts are not added.
Relatively easy and inexpensive to install at existing facilities	Requires tertiary filtration to remove P in suspended solids.
Sludge can be processed in the same manner as in non-P-removal systems	Coloured effluents if iron salts are used.
Primary clarifier metal addition can reduce organic load to secondary unit by 25–35%.	Toxicity problems if the process is handled inefficiently.
Biological P removal	
The amount of sludge generated is comparable to conventional activated sludge systems.	In all but the PhoStrip[b] system, P removal performance is controlled by the BOD:P ratio of the wastewater
Can be installed at existing plug-flow[c] activated sludge plants with little or no equipment changes or additions, provided that the plant has sufficient capacity.	Highly efficient secondary clarifier performance is required to achieve an effluent concentration of 1 mg P l^{-1}.
Existing sludge handling equipment can be used at retrofitted plants, provided phosphorus is not solubilised and returned to the plant.	Not easily retrofitted into fixed-film biological systems.
Little or no chemical costs nor handling equipment, except for PhoStrip process2, metal salt supplement or for effluent polishing.	Potential for P release in sludge handling system.
With some processes, P removal can be achieved together with nitrogen removal at virtually no additional operating cost.	Recycle streams have low P content.
Better control of filamentous organisms is possible in some systems.	Standby chemical feed equipment may be required in case biological P removal efficiency falls.
	Sludge is often poor-settling.
	Requires increased retention time in activated sludge plants.

[a] Pickle liquors — waste product from metal processing that contains iron salts.
[b] The Phostrip process — commonly used biological phosphorus stripping process.
[c] Plug-flow — whereby BOD loading is delivered to the plant in discrete plugs associated with domestic behaviour.

needs to be fully considered, including various pieces of existing legislation and possibilities under the Water Framework Directive for bringing in new regulatory measures that are focused specifically on the diffuse pollution problem.

10. Control of point sources

STW outfalls are the major point source inputs to rivers, with the phosphorus originating primarily from human waste and detergents but also trade wastes and other ingressions to the sewerage system. After secondary treatment at the works, phosphorus is largely present as bioavailable orthophosphate, which reacts and precipitates out of solution in the presence of metal salts, can be taken up by micro-organisms or isolated from the waste stream by magnetic or ion exchange processes. A number of options, therefore, exist for stripping phosphorus from waste waters, within which lie a large number of variations. The two most widely used removal methods in the UK and overseas are chemical precipitation (by the addition of metal salts) and biological removal (Day and Cooper, 1992). In the future, other processes such as magnetic removal and ion exchange may be developed to a stage where they can be used operationally (Morse et al., 1998).

The advantages and disadvantages of chemical and biological treatments are shown in Table 6. In general terms, biological treatment is a more environmentally benign process in that sludge production is not increased (merely enriched) and there is no consumption/disposal of metal ions to consider. However, it cannot achieve the same phosphorus removal efficiencies as chemical treatment and is currently rather variable in performance without chemical supplements. This said, chemical treatment only achieves enhanced removal efficiencies (below 2 mg l^{-1}) at the expense of producing large amounts of extra sludge.

It should be noted that biological removal is complex and water companies are only likely to favour its installation where the STW is sufficiently large for labour to be available for monitoring its performance (generally above 50 000 pe). However, on-line colourimetric methods exist for monitoring phosphorus concentrations in real time after floc settlement, whereupon chemical salts can be automatically added as required to make up for the variable biological performance. In well-run modern plants, only occasional chemical additions may be required to keep effluent concentrations under 2 mg l^{-1} (CEEP, 1998). In addition, removal is generally more consistent in summer (due to greater consistency in influent wastewater), a time when phosphorus loads from STWs are most critical ecologically.

Ultimately, the choice between the two approaches is site- and cost-dependent. Biological removal requires an activated sludge plant, which is costly to install if not already in place at the works. In general, activated sludge is only used at larger treatment works, and so these are more amenable to biological removal (approx. 200 plants over 10 000 pe have activated sludge plants in the UK). Most smaller plants only have conventional secondary treatment, and so the only on-site treatment option is some form of chemical dosing. Of the chemical processes, calcium dosing to produce calcium phosphates is probably the best environmental option since the resultant sludge is of higher agricultural value (where dosing occurs within the secondary treatment plant) and it can also be recovered for agro–industrial applications (if dosed into the effluent after secondary treatment), particularly for use as inorganic fertiliser. Calcium dosing and appropriate management of magnesium and ammonium concentrations can produce struvite (a mixture of magnesium ammonium phosphates and potassium ammonium phosphates), another material suitable for recovery for agro–industrial use.

An important calcium-based technique is a Dutch process called the Crystalactor. After removing carbon dioxide by degassing, the effluent is dosed with lime to precipitate calcium phosphate, with sand particles acting as the seeding agent for pellet growth. The pellets can be dried to below 5–10% water and contain 5–15% phosphorus (compared to approx. 20% in rock phosphate). A few full-scale plants are in operation in the Netherlands, whilst operational plants based on similar processes or struvite formation exist in Japan (CEEP, 1998). Phosphorus recovery is in-

evitably more economically viable at the largest works, but it remains to be seen how small a works can be before the approach becomes impractical. Importantly, the cost of establishing phosphorus-stripping can be partially offset by the sale of recycled phosphorus to industry, who can be expected to pay a small premium for the product compared to rock phosphate due to its relative purity.

In the UK, the majority of sewage treatment works incorporate primary (solids separation) and secondary (typically biological filtration) treatment. Whilst there are notable exceptions, such as the installation of P-stripping at all of the major STWs feeding Lough Neagh during the 1970s, tertiary treatment to remove phosphorus has been rare. The primary driver for the installation of P-stripping in the UK has until recently been the Urban Waste Water Treatment Directive. 'Sensitive areas' have to be identified by Member States under the Directive, which relate to waters that are, or are in danger of becoming, eutrophic. Within sensitive areas, 'appropriate' treatment for nitrogen and/or phosphorus removal has to be installed at all works serving more than 10 000 population equivalents (the issue is generally phosphorus in freshwater). In the first round of identifying 'sensitive areas', 37 sites (including 23 river stretches) were designated within the UK as requiring the installation of P-stripping, constituting a small proportion of the total sewage effluent load generated in the UK (some 5.2%; IEEP, 1999). In some of the designated catchments all major works were already equipped with P-stripping. Further designations have been made, but the revised list still constitutes a small proportion of the total sewage effluent load. Certainly, P-stripping can significantly reduce phosphorus concentrations in rivers, but not always to levels where biological improvement occurs (Jarvie et al., 2002b; Neal et al., 2002).

The criteria for designating sensitive waters are set on a national basis so there is plenty of scope for interpretation. The approaches of other member states vary, but all of those in north-west Europe have taken a much stricter stance than the UK towards phosphorus control. In Denmark, the Netherlands, Sweden and Finland, the whole territory has been designated as sensitive area, whilst the majority of Germany and a large proportion of France have also been designated (IEEP, 1999). The result is that phosphorus removal will be undertaken at nearly all works in north-west Europe that are over 10 000 pe.

The approaches of most of these north-western European countries to phosphorus removal go considerably further than required by the Directive. Germany, Denmark, the Netherlands and Sweden have very strict controls on phosphorus loads in effluents and have spent considerable effort on removing phosphorus from smaller works. Germany had approximately 900 plants removing phosphorus by 1999, compared to 23 plants operating in the UK by 1997 (IEEP, 1999). Since these countries have been actively seeking phosphorus removal for some years, existing plants tend to utilise chemical removal with iron salts. However, practices are changing as new information and technology comes to light; for instance, the policy in Denmark has changed from iron-dosing to biological treatment where possible.

Within the past 2 years, the UK stance on phosphorus removal has moved towards a more ecologically-based and precautionary strategy, focused on the protection of freshwater sites designated for nature conservation. The Government has laid out plans to introduce phosphorus removal at up to 74 sewage treatment works affecting freshwater Sites of Special Scientific Interest (SSSIs), to be implemented by 2005 through the water industry's Asset Management Planning programme (DETR, 1998).

11. Internal loading of phosphorus from riverine sediments

In rivers where inputs have been high for a long period of time, a reservoir of phosphorus can accumulate in the bed sediments that can continue to contribute to the nutrient status of the river after inputs have been reduced. This phosphorus can be taken up by plants directly from the sediment (via the roots of higher plants or through uptake by benthic algae), or following

release into the water column (via the shoots of rooted plants or through uptake by epiphytic or epilithic algae). Importantly, the lower the target concentration set for phosphorus, the more influential internal sources are likely to be. The internal phosphorus reservoir is most likely to cause problems in sluggish river reaches with silty bed sediments (i.e. depositional zones), which provide a trap for phosphorus-rich particulates and a large capacity for adsorbing soluble phosphorus from the water column. However, river sections with coarser substrates that are suffering from siltation problems are also of major concern. Importantly, anoxic conditions can occur in silty substrates, which enhances the release rate of phosphorus (Mainstone et al., 1996) in addition to having direct impacts on the establishment, growth and function of the roots of higher plants.

Depending on local circumstances, it may be appropriate to dredge the surface layers of sediments out of depositional reaches of the river, in order to reduce the size of the internal reservoir. In gravels silted up with phosphorus-enriched silt, it may be useful to artificially clean the substrate (through raking or jet-hosing). Such action should be decided after an assessment of the importance of internal loading to the river in question, which would sensibly be based on measurements of the EPC and redox potential of the sediment. It should be remembered that dredging can have detrimental effects on instream habitats and can also reduce the capacity of the river to provide water to floodplain habitats. In addition, downstream habitats may be severely affected by heavy loads of silt washed out of upstream areas. Care should, therefore, be taken in the design of dredging operations to minimise ecological impacts. Finally, it is important that phosphorus inputs to the river are tackled holistically so that further accumulation is avoided — both point sources and diffuse sources contribute to the internal reservoir and, therefore, need to be addressed in combination.

12. Catchment-level fluxes of phosphorus

Phosphorus recycling for agro–industrial use has the potential to replace a non-renewable resource with a sustainable reclaimed resource, benefiting the water industry in economic and regulatory contexts, and reducing the mass-transfer of phosphorus into UK catchments (thereby tackling eutrophication problems at their true source). Active encouragement of this approach is likely to bring considerable environmental benefits.

Currently, UK catchments generally receive large net imports of phosphorus due in large part to the agricultural use of considerable amounts of imported rock phosphate fertiliser. It is essential that these phosphorus imports are reduced if we are to control accumulation in catchment soils and losses to the river network. Using fertilisers in line with crop and soil requirements is the most important consideration, but in addition, every effort should be made to ensure that the phosphorus removed from effluent streams is properly utilised within agricultural systems and thereby replaces an equivalent amount of imported P. This requires the use, wherever possible, of biological removal processes or chemical removal processes based on calcium.

Phosphorus recovered from STWs is potentially highly attractive to industry compared to rock phosphate (the current raw material used), generally having lower concentrations of troublesome contaminants (particularly heavy metals). In scavenging heavy metals from the wastewater stream, the recovery process reduces the metals load from STW with no disbenefit to industry (recovered phosphorus still contains levels of metals one or two orders of magnitude lower than rock phosphate — CEEP, 1998).

13. Making management decisions at a catchment level

The steps involved in developing a phosphorus control programme at a catchment level are outlined in Fig. 7. In general, point sources (and in particular sewage treatment works) can be expected to be a high priority in control programmes even if their contribution to the annual phosphorus budget is relatively small, represent-

ing discrete and highly treatable sources that provide very bioavailable phosphorus to the water column at a time of minimum dilution and maximal plant activity. Diffuse loads need to be assessed in relation to their incorporation into riverine sediments and their subsequent utilisation by the plant community, as well as any potential for chronic leaching of phosphorus into groundwaters and, hence, baseflows. In terms of control, diffuse loads need to be evaluated at

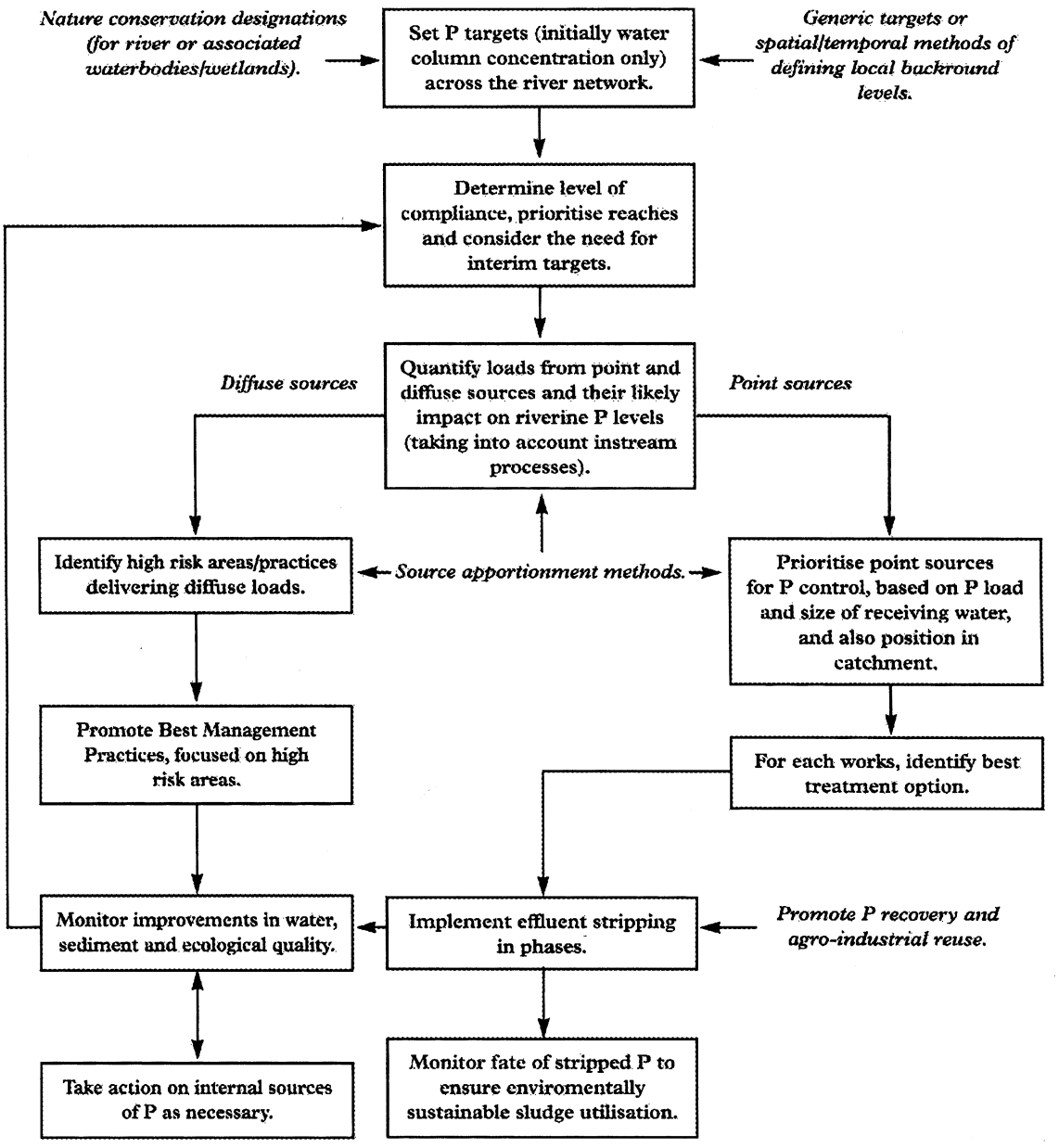


Fig. 7. Process diagram for phosphorus control within a river catchment.

sufficient resolution to identify high risk areas within the catchment, using modelling techniques but also including field investigations of run-off pathways and land management (in relation to factors such as the size of the soil phosphorus reserve and levels of erosion risk), so that resources invested in the promotion and implementation of best practices has greatest benefits.

Even the smaller point sources need to be considered for treatment since they can have significant effects in combination, particularly in headwater areas where relatively small loads can induce ecologically relevant enrichment. The largest works will not always be the most critical in relation to establishing phosphorus-stripping; the position in the catchment is important, and it may be that removing phosphorus from a smaller works upstream will have greater benefits to the river as a whole than treating a larger works in the lower reaches. When phasing phosphorus removal at a range of works within a catchment, it makes sense ecologically to focus on headwater areas and work downstream, assessing chemical and biological benefits along the way.

Table 7 gives a summary of the management options for removing phosphorus from effluents, indicating the environmental desirability of each. The appropriate strategy for smaller point sources needs to be considered carefully in relation to local circumstances. There may be opportunities for transferring effluents from smaller works to larger works where phosphorus removal is more cost-effective and more environmentally desirable processes can be used (particularly in relation to recovery for agro–industrial reuse). However, these smaller works typically return waters to the river that have been abstracted within the catchment, such that transfer downstream to larger works can extend the zone of hydrological impact. One possibility for avoiding adverse effects is to pump stripped effluent back up the catchment, but the feasibility of such mitigation measures will vary greatly between works.

14. Final thoughts

The complex nature of competition within plant communities and the interaction of nutrient enrichment with many other anthropogenic impacts means that the ecological benefits accruing specifically from a nutrient reduction programme cannot be stated with any certainty. This requires that a pragmatic, precautionary approach is taken to developing strategies for phosphorus control, focused on ecological risk rather than observed impacts and measurable ecological benefits. The momentum now generated for the control of domestic point sources needs to be maintained by the identification and treatment of further problem discharges (including industrial effluents), and urgently needs to be extended to the control of

Table 7
Guide to the environmental desirability of different phosphorus-stripping treatments

Treatment type	Without P recovery	With P recovery
Biological treatment	8	10
Calcium phosphate/struvite recovery	N/A	10
Other novel P recovery techniques (e.g. ion exchange)	N/A	Evaluate method individually
Where the above are not feasible (due to works size / capacity)		
Re-routing effluent to larger works where above options are in place or viable[a]	8	10
Calcium dosing	6	10
Iron dosing	4	Not possible
Aluminium dosing	0	Not possible

Numbers are subjective ratings on a 10-point scale, with 10 representing the most desirable options.
[a]Care needs to be taken over potential hydrological effects associated with moving points of effluent discharge.

diffuse sources so that ecological benefits can be maximised.

The need for such an approach is nowhere more apparent than for Special Areas of Conservation designated under the EC Habitats Directive, where the precautionary approach is enshrined in legislation. Positive management of these sites should act as a template for the implementation of control strategies elsewhere.

References

Billen G, Lancelot C, Meybeck M. N, P, and Si retention along the aquatic continuum from land to ocean. In: Mantoura FFC, Martin JM, Wollast R, editors. Ocean margin processes in global change. John Wiley and Sons Publishers, 1991:19–44.

CEEP. Recovery of phosphorus for recycling from sewage and animal wastes: summary of conclusions and discussions. International conference organised by the Centre Europeen d'Etudes des Polyphosphates at Warwick University 6–7 May, 1998:1998.

Chambers PA, Prepas EE. Nutrient dynamics in riverbeds: the impact of sewage effluent and aquatic macrophytes. Water Res 1994;28(2):453–464.

Day M, Cooper PF. Guide on nutrient process selection. Foundation for Water Research report No. FR 0253. Medmenham: WRc, 1992.

DETR. Raising the quality: guidance to the Director General of Water Services on the environmental and quality objectives to be achieved by the water industry in England and Wales, 2000–2005. Department of the Environment, Transport and the Regions, 1998.

Environment Agency. A price worth paying. Bristol: Environment Agency, 1998.

Environment Agency. Aquatic eutrophication in England and Wales: a management strategy. Bristol: Environment Agency, 2000, p. 32.

Hartley AM, House WA, Callow ME, Leadbetter BSC. Coprecipitation of phosphorus with calcite in the presence of photosynthesizing green algae. Water Res 1997;31: 2261–2268.

Heathwaite AL. Sources and pathways of phosphorus loss from agriculture. In: Tunny H, Carton OT, Brookes PC, Johnston AE, editors. Phosphorus loss from soil to water. CAB International, 1997:205–223.

House WA, Denison FH. Nutrient dynamics in a lowland stream impacted by sewage effluent — Great Ouse, England. Sci Total Environ 1997;205(1):25–49.

House WA, Denison FH, Armitage PD. Comparison of the uptake of inorganic phosphorus to a suspended and stream bed-sediment. Water Res 1995;29:767–779.

Institute of European Environmental Policy — IEEP, The implementation of the 1991 EU Urban Waste Water Treatment Directive and its role in reducing phosphate discharges – summary of report. SCOPE Newsletter No. 34, October 1999.

Iversen TM, Kjeldsen K, Kristensen P, de Haan B, van Oirschot M, Parr W, Lack T. Integrated environmental assessment on eutrophication. A pilot study. NERI Technical Report 207. Denmark: National Environmental Research Institute, 1997.

Jarvie HP, Neal C, Warwick A, White J, Neal M, Wickham HD, Hill LK, Andrews MC. Phosphorus uptake into algal biofilms in a lowland chalk river. Sci Total Environ 2002a;282–283:353–373.

Jarvie HP, Lycett E, Neal C, Love A. Patterns in nutrient concentrations and biological quality indices across the upper Thames basin, UK. Sci Total Environ 2002b;282–283:263–294.

Johnes P, Moss B, Phillips G. Lakes — classification and monitoring. A strategy for the classification of lakes. NRA R&D Note 253, NRA Project Record 286/6/A, 1994.

Kilham P, Hecky RE. Comparative ecology of marine and freshwater phytoplankton. Limnol Oceanogr 1988;33: 779–795.

MAFF. The code of good agricultural practice for the protection of water. London: HMSO, 1998, p. 97.

MAFF. Controlling soil erosion: a manual for the assessment and management of agricultural land at risk of water erosion in lowland England. London: Ministry of Agriculture, Fisheries and Food, 1999, p. 44.

Mainstone CP, Holmes NT, Armitage PD, Wilson AM, Marchant JH, Evans K, Solomon D. Chalk rivers-nature conservation and management. WRc Report to English Nature and the Environment Agency. Medmenham: WRc, 1999a, p. 184.

Mainstone CP. Cooperative agreements in agriculture: UK report. Environment Agency Draft R&D Technical Report. Medmenham: WRc, 1999b, p. 40.

Mainstone CP, Davis RD, House A, Parr W. A review of methods for assessing and controlling non-point sources of phosphorus. NRA R&D Project Record 562/5/W. Bristol: NRA, 1996, p. 169.

Mainstone CP, Ashley S, Gunby A, Parr W, Woodrow D, Turton P, McAllem Y. Development and testing of General Quality Assessment Schemes: Nutrients in rivers and canals. NRA R&D Project Record 469/11/HO. Bristol: NRA, 1995, p. 172 plus appendices.

Morse GK, Brett SW, Guy JA, Lester JN. Review: phosphorus removal and recovery technologies. Sci Tot Environ 1998;212:69–81.

Morse G, Eatherall A, Jenkins A. Managing agricultural pollution using a linked geographical information system and non-point source pollution model. J Inst Water Environ Manage 1994;8:277–286.

Moss B. The restoration of freshwater ecosystems — an overview. River Restoration'96 — plenary lectures. Proceedings of an international conference arranged by the European Centre for River Restoration. Denmark: National Environmental Research Institute, 1997.

Mulholland PJ, Hill WR. Seasonal patterns in streamwater nutrient and dissolved organic carbon concentrations: separating catchment flow path and instream effects. Water Resour Res 1997;33(6):1297–1306.

Neal C. Interception and attenuation of atmospheric pollution in a lowland ash forested site, Old Pond Close, Northamptonshire, UK. Sci Total Environ 2002a;282–283:99–119.

Neal C. Calcite saturation in eastern UK rivers. Sci Total Environ 2002b;282–283:311–326.

Neal C, Jarvie HP, Howarth SM, Whitehead PG, Williams RJ, Neal M, Harrow M, Wickham H. The water quality of the River Kennet: initial observations on a lowland chalk stream impacted by sewage inputs and phosphorus remediation. Sci Total Environ 2000a;251/252:477–495.

Neal C, Williams RJ, Neal M, Bhardwaj LC, Wickham H, Harrow M, Hill LK. The water quality of the River Thames at a rural site downstream of Oxford. Sci Total Environ 2000b;251/252:441–457.

Neal C, Jarvie HP, Wade AJ, Whitehead PG. Water quality functioning of lowland permeable catchments: inferences from an intensive study of the River Kennet and upper River Thames. Sci Tot Environ 2002;282–283:3–7.

Parr W, Mainstone CP. The Pevensey Levels-data to support its inclusion as a candidate eutrophic sensitive area. Unpublished report to Environment Agency Southern Region, 1997.

Parr W, Andrews K, Mainstone CP, Clarke SJ. Diffuse pollution: sources of N and P. WRc Report to the Department of the Environment, Transport and the Regions. DETR Report No. DETR 4755. Medmenham: WRc, 1999:162.

Parr W, Mainstone CP, Morgan N. Nutrient budget for the upper reaches of the Hampshire Avon. Report to English Nature and the Environment Agency. Medmenham: WRc, 1998.

Pelton DK, Levine SN, Braner M. Measurements of phosphorus uptake by macrophytes and epiphytes from the Laplatte River using P-32 in stream microcosms. Freshwater Biol 1998;39(2):285–299.

Reynolds CS. The ecology of freshwater phytoplankton. Cambridge University Press, 1984.

Robach F, Merlin S, Rolland T, Tremolieres M. Ecophysiological approach of water quality bioindication using aquatic plant materials. Ecologie 1996;27(4):203–214.

Robach F, Hajnsek I, Eglin I, Tremolieres M. Phosphorus sources for aquatic macrophytes in running waters: water or sediment? Acta Botanica Gallica 1995;142(6):719–731.

Scheffer M, De Redelijkheid MR, Noppert F. Distribution and dynamics of submerged vegetation in a chain of shallow eutrophic lakes. Aquat Botany 1992;42:199–216.

Timms RM, Moss B. Prevention of growth of potentially dense phytoplankton populations by zooplankton grazing, in the presence of zooplanktivorous fish, in a freshwater wetland ecosystem. Limnol Oceanogr 1984;29:472–486.

Vighi M, Chiaudani G. A simple method to estimate lake phosphorus concentrations resulting from natural background loadings. Water Res 1985;19:987–991.